RAMAKRISHNA MISSION VIDYAMANDIRA

(Residential Autonomous College affiliated to University of Calcutta)

B.A./B.Sc. SIXTH SEMESTER EXAMINATION, JUNE 2022

THIRD YEAR (BATCH 2019-22)

COMPUTER SCIENCE (HONOURS)

Paper : XIII [CC13]

Date : 09/06/2022

Time : 11.00 am – 1.00 pm

Full Marks : 50

Answer **any five** of the following questions : [5×10]

1. a) Consider the following states of a puzzle:

Initial State :

1	2	3
	4	6
7	8	5

Goal State :

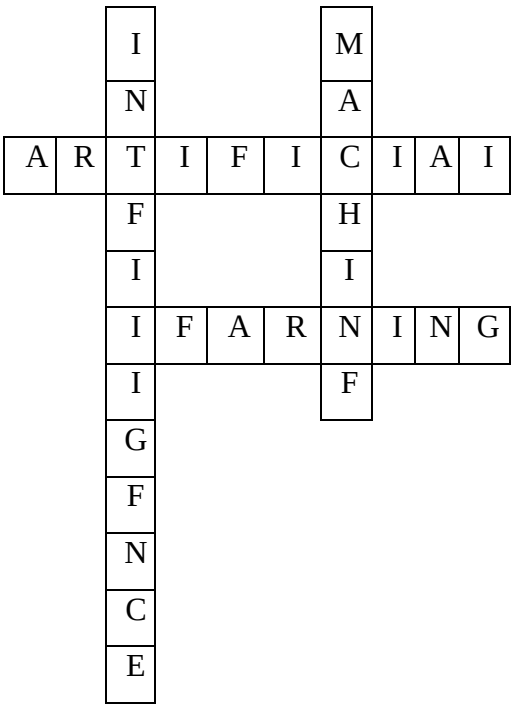
1	2	3
4	5	6
7	8	

Show how traditional hill climbing approach will struggle to find a solution while using the heuristic 'Number of misplaced tiles'.

b) What do you mean by 'PEAS' for an intelligent agent?

c) Define State Space search. (4+4+2)

2. a) Consider the following crossword puzzle problem. In this variant, we assume that we have a set of words and a crossword puzzle grid. Our goal is to fill the crossword grid with the words such that letters of intersecting words match.



Provide a constraint satisfaction problem formulation for this variant of the crossword puzzle problem.

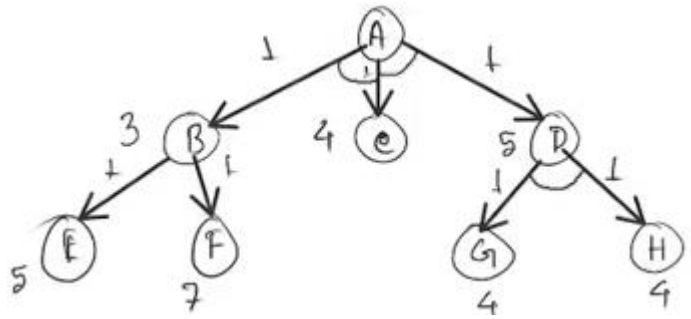
i) Specify the variables to which values are to be assigned.

ii) Specify the domains from which the variables take their values.

iii) Define the constraints that must hold between variables.

b) How can performance of depth first search be improved? (6+4)

3. a) Perform AO* search in the following graph:



b) What do you mean by 'Metaheuristic' search?

c) How α-β pruning helps improving performance of Minimax? (4+3+3)

4. a) Write the following sentences in FOL and then covert them to Normal forms:

i) Every Philosopher writes at least one book.

ii) Every dog owner is an animal lover.

b) Write a short note of Stochastic Hill Climb algorithm. (6+4)

5. a) Consider the following set of axioms:

i) Everyone who loves someone who loves them back is happy.

ii) Mary loves everyone.

iii) There is someone who loves Mary.

From the above statements conclude that 'Mary is Happy'.

b) What do you mean by Skolemization? Give example. (6+4)

6. a) Write short note on components of an Expert Systems (ES).

b) In which cases Prolog generates redundant inference and Infinite loops?

c) "Prolog uses backward chaning for inference" –Explain. (5+3+2)

7. a) Define Core and Support of a fuzzy set with suitable examples.

b) Explain any two types of fuzzy membership functions and give practical examples of their use case scenarios.

c) Why is 'Soft Computing' different from traditional computing approach? (4+4+2)

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